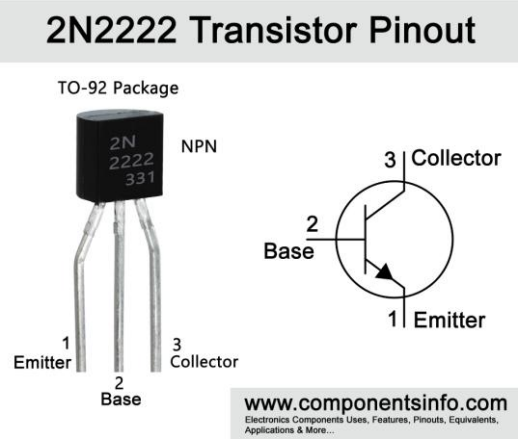
First Transistor, December 1947

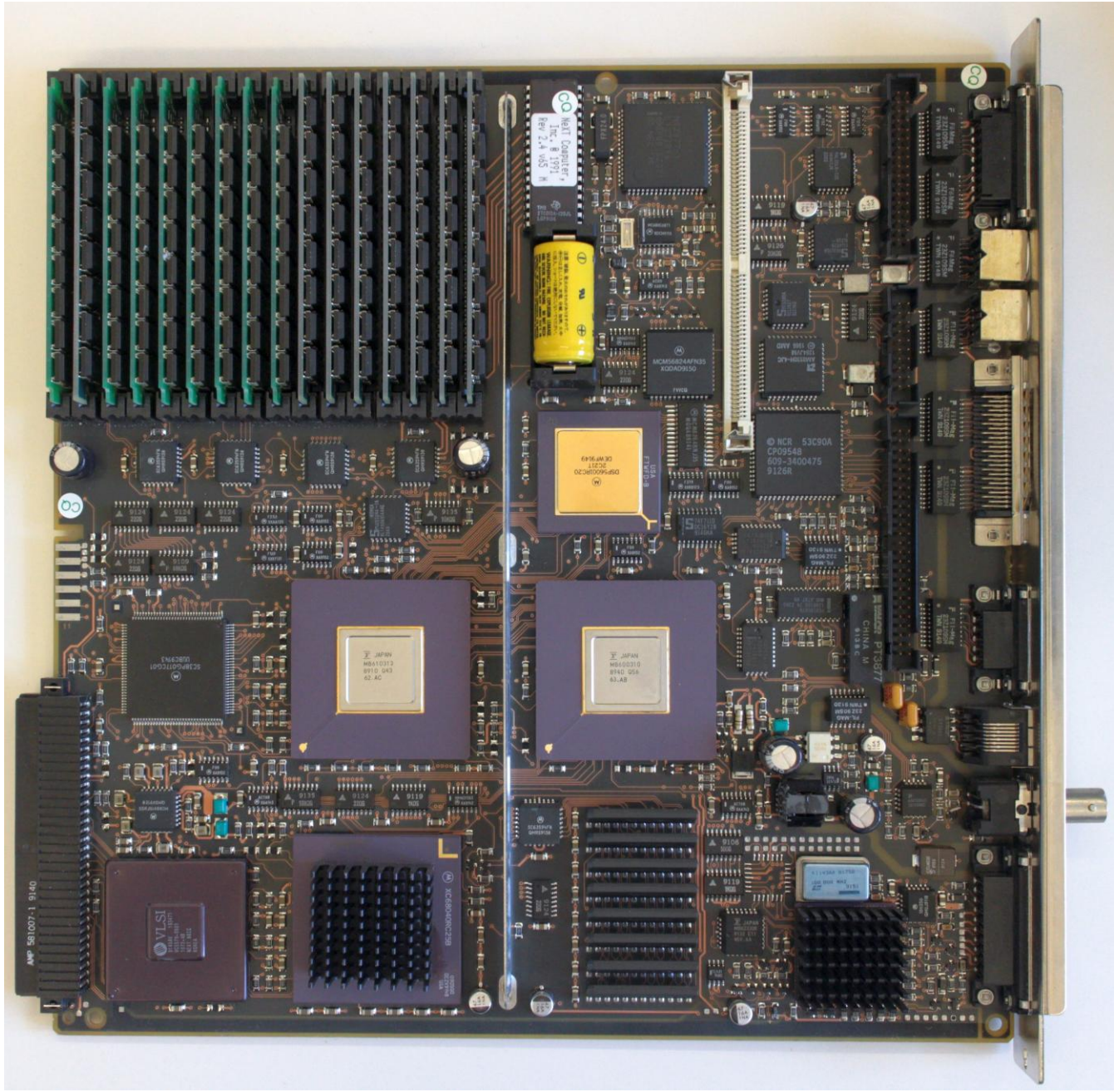
John Bardeen, William Shockley and Walter Brattain



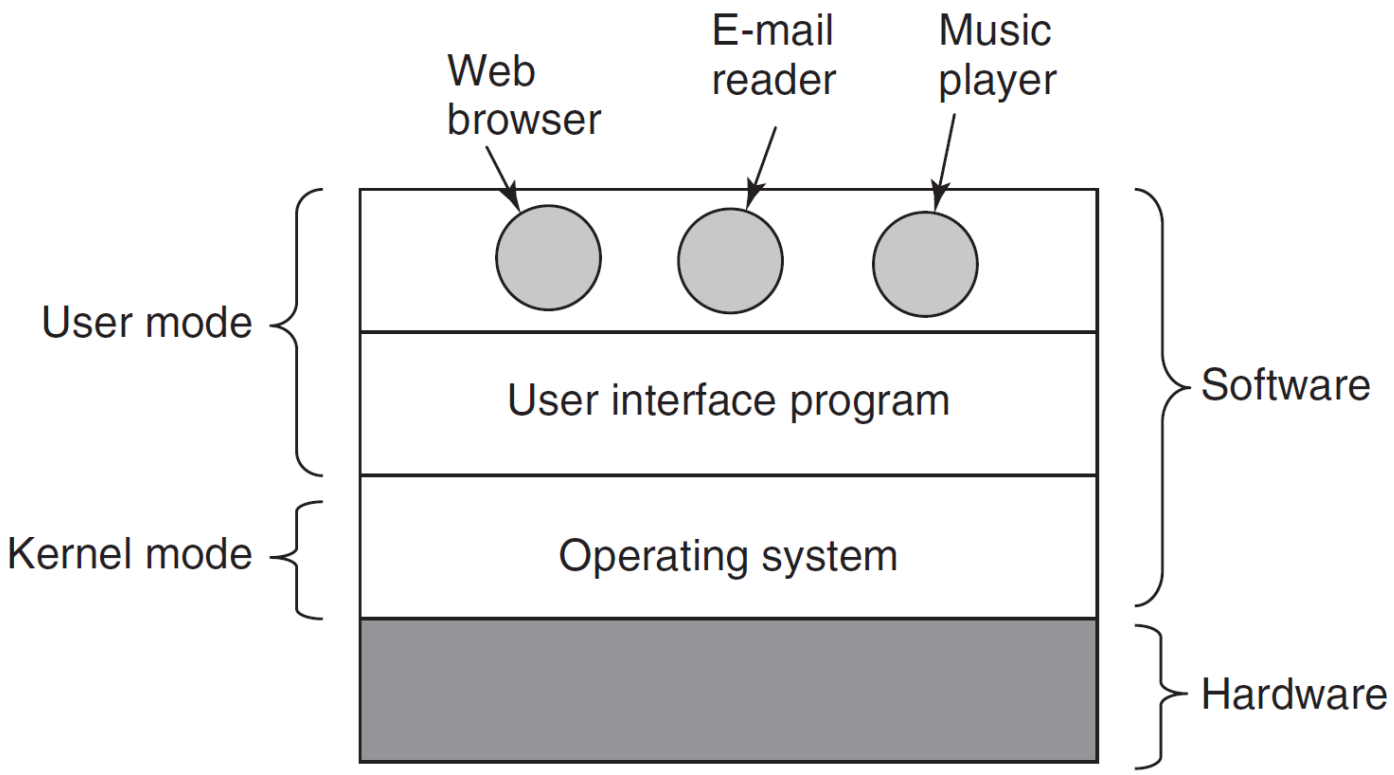
NeXTcube Workstation, 1990

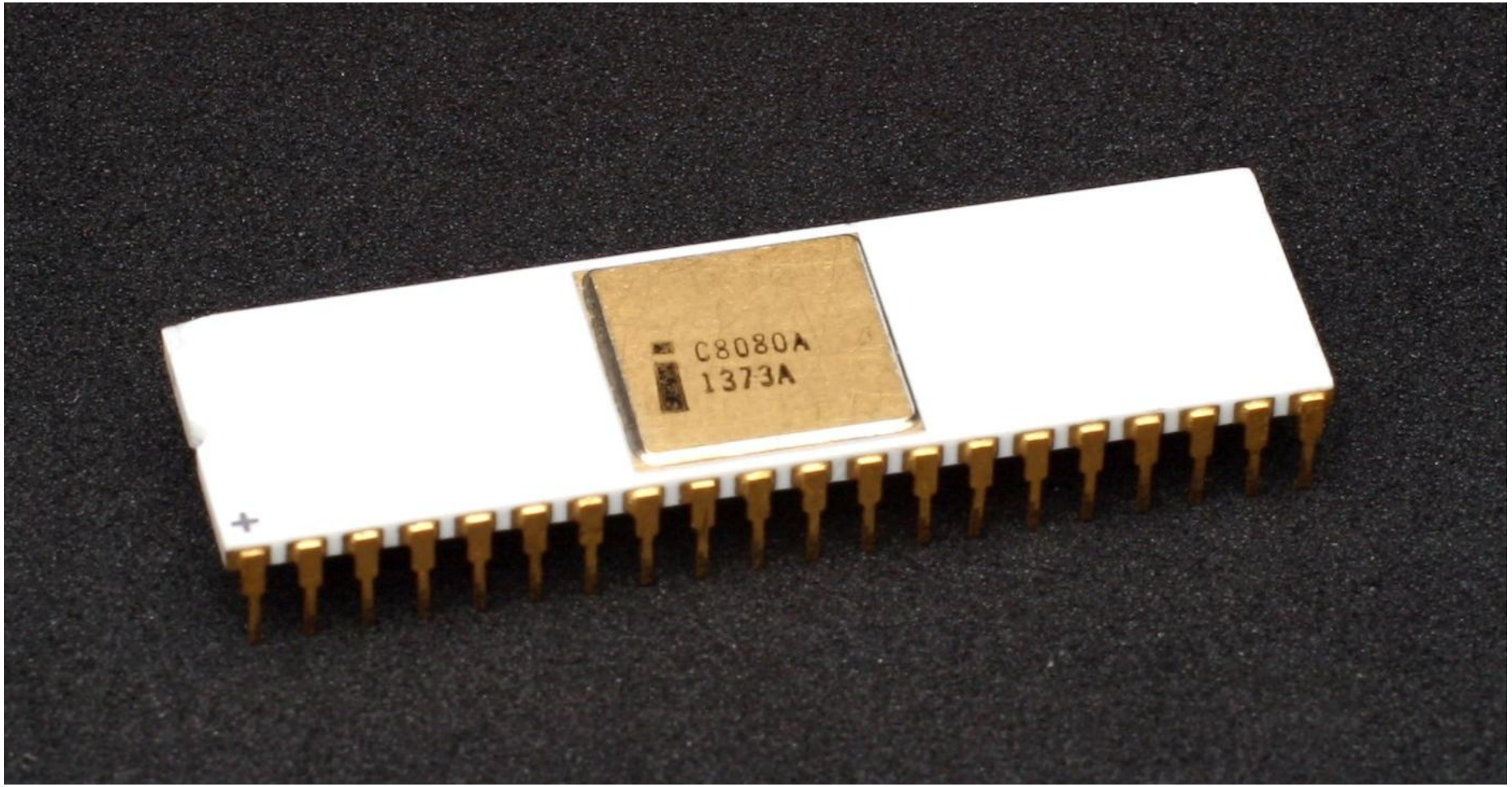
A Motorola 68040 microprocessor
40MHz, 32 bits, 12.M Transistor

NeXTcube Workstation, 1990



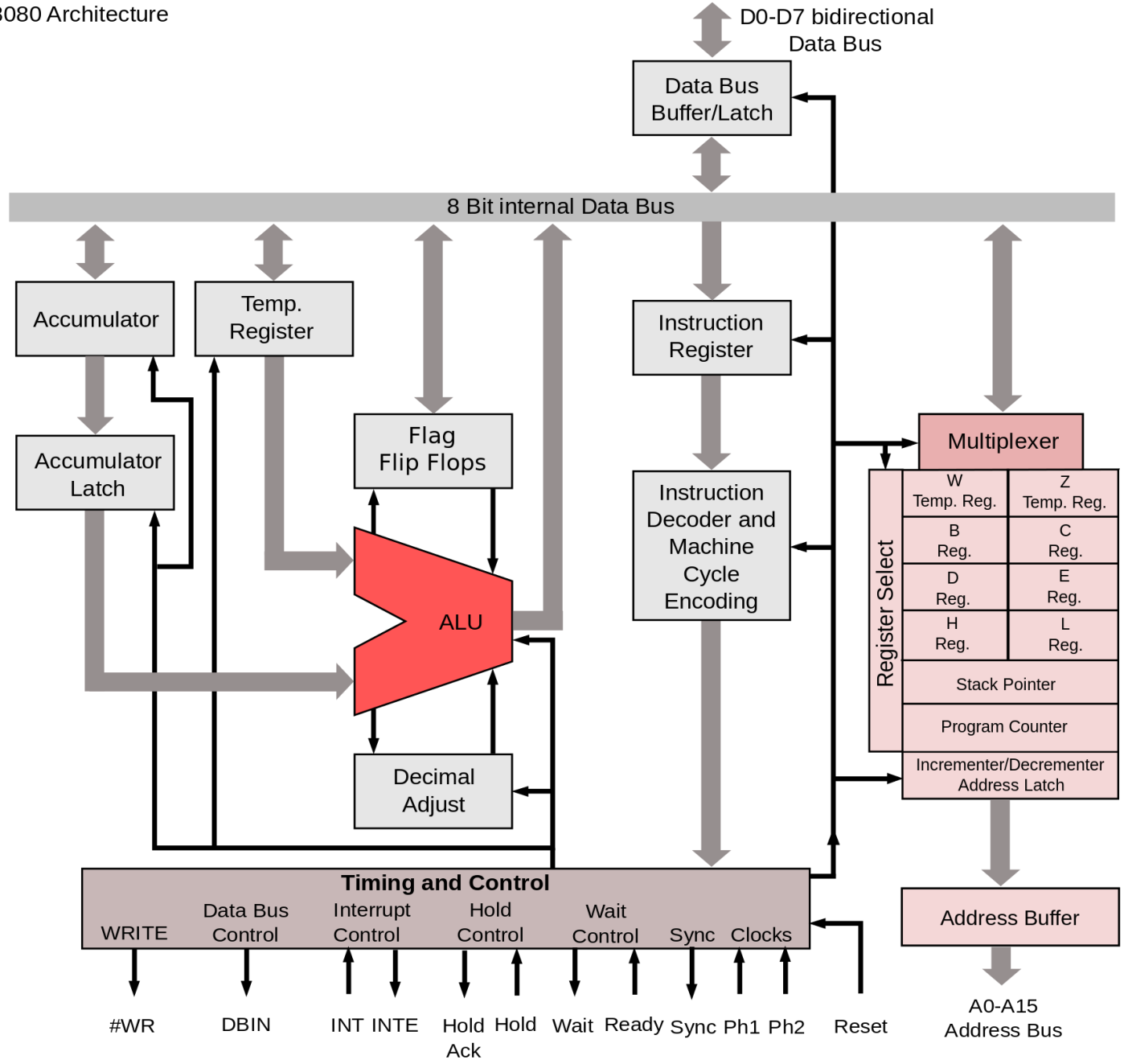
REPUBLIC OF GAMERS EST. 2006
CROSSHAIR
AMD
RYZEN
AM5
START
FLEXKEY
ROG
EXTREME
SUPREME FX


Web browser
E-mail reader
Music player
User mode
User interface program
Software
Kernel mode
Operating system
Hardware



Intel 8080: 4,500 Transistors, 6um , 8-Bit, Clock Rate: 2MHz, April 1974

Intel 8080 Architecture

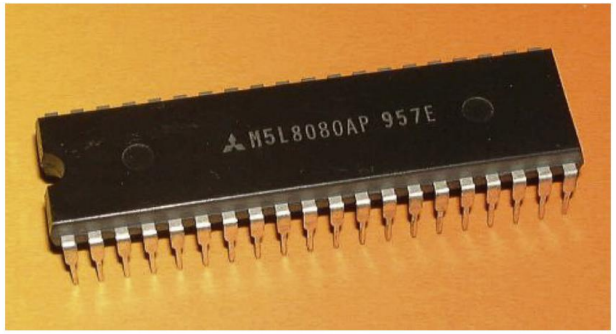


April 1974, Clock Rate, 2 MHz to 3.125 MHz, 4500 Transistors, 6um, 8 Bit

AMD Am9080

CEMI MCY7880 (Poland)

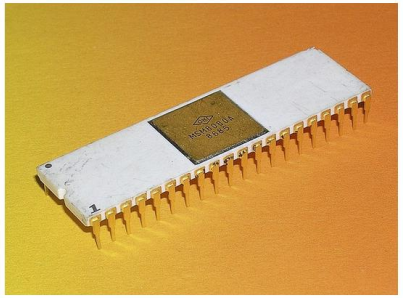
Kvazar Kiev K580IK80 (Soviet Union)



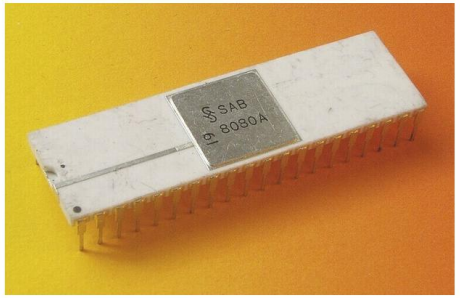
Mitsubishi Electric M5L8080

National Semiconductor INS8080

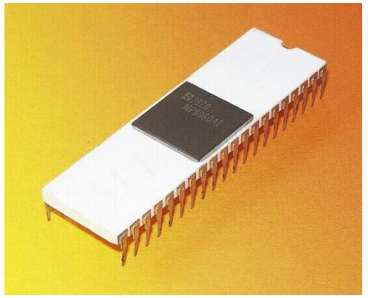
NEC μPD8080AF



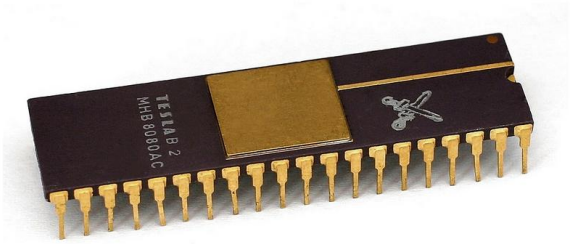
OKI MSM8080



Siemens SAB8080



Signetics MP8080

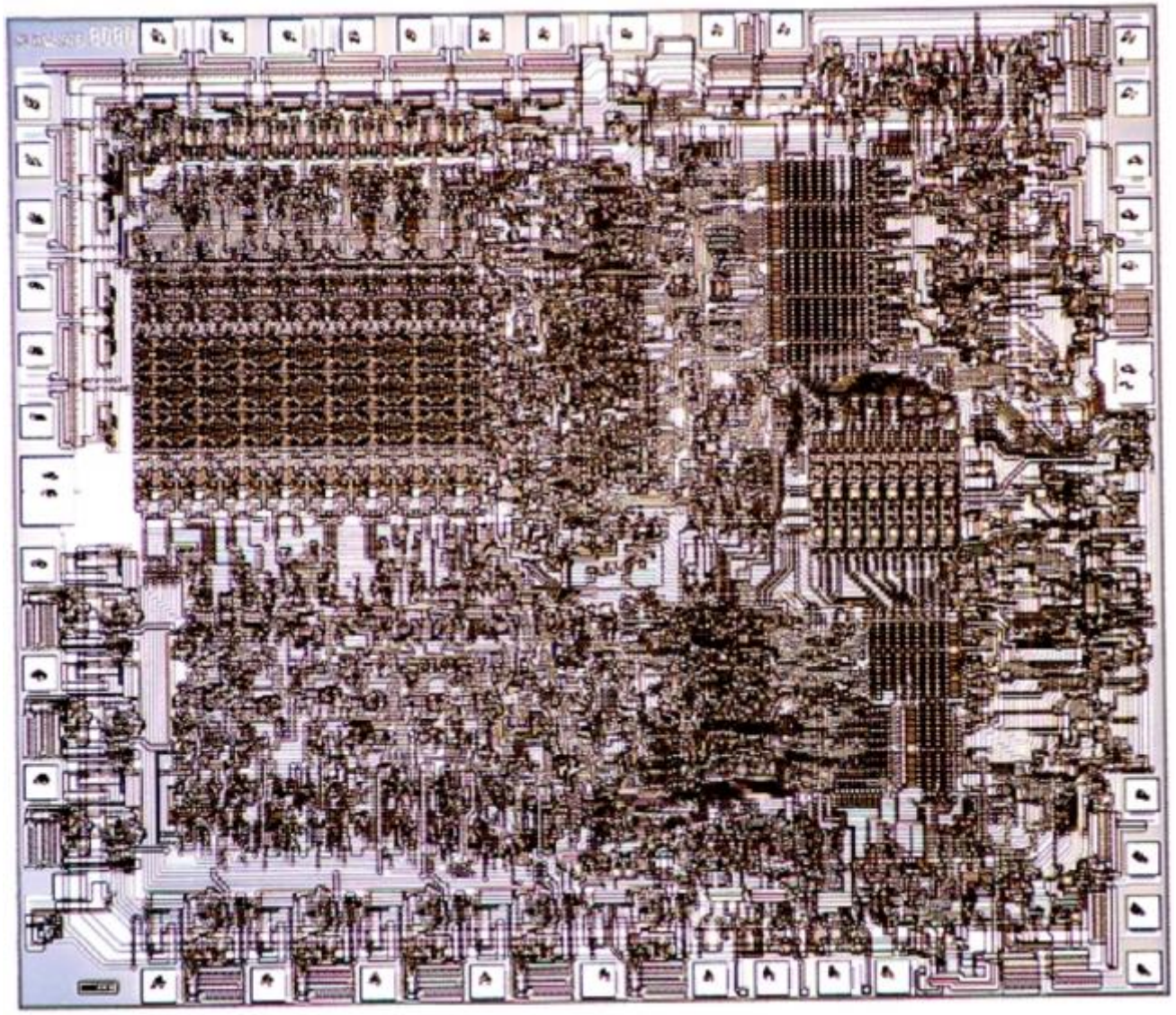


Tesla (Czechoslovak company) MHB8080

Texas Instruments TMS8080

As of 2017, the 8080 is still in production at Lansdale Semiconductors

April 1974, Clock Rate, 2 MHz to 3.125 MHz, 4500 Transistors, 6um, 8 Bit



6 µm process
nMOS transistors
4,500 transistors
0.164" x 0.19"
20.1 mm² die size

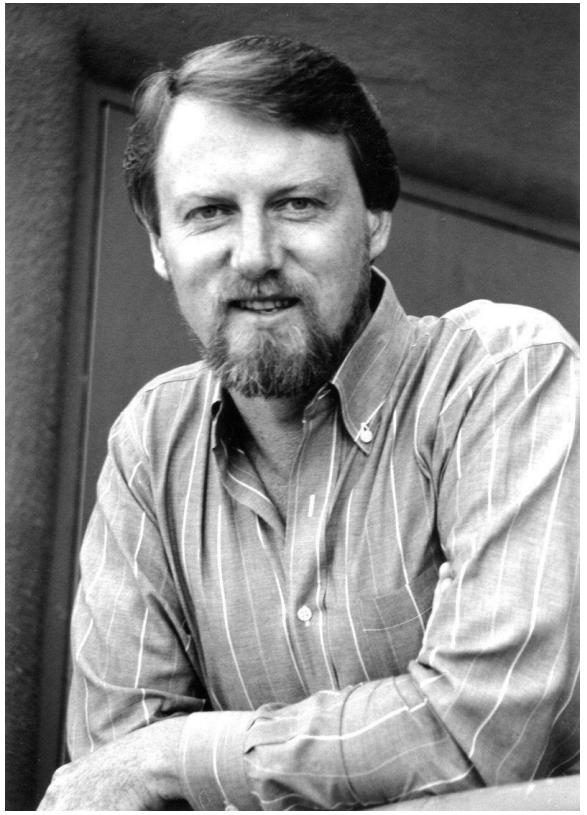
Pioneer Operating Systems

Gary Kildall developed the first high-level programming language for microprocessors, PL/M, in 1973.

In 1974, Kildall OS on Intel 8080, Gary Kildall, disk-based OS called CP/M (Control Program for microcomputers) to enable the 8080 to control a floppy disk drive, combining for the first time all the essential components of a computer at the microcomputer scale. He demonstrated CP/M to Intel, but Intel had little interest and chose to market PL/M instead

In 1977, Digital Research adapted CP/M for many microcomputers, 8080, Zilog Z80 etc. (pioneer of BIOS)
Pazarı 5 yıl boyunca elinde tuttu.

In 1980s, DOS (Disk Operationg System) led by IBM (IBM PC, BASIC Interpreter), Tim Peterson (Microsoft). Later developments named as MS-DOS (MicroSoft Disk Operating System, written x86 assembly). Later microprocessors, Intel 80386, 80486



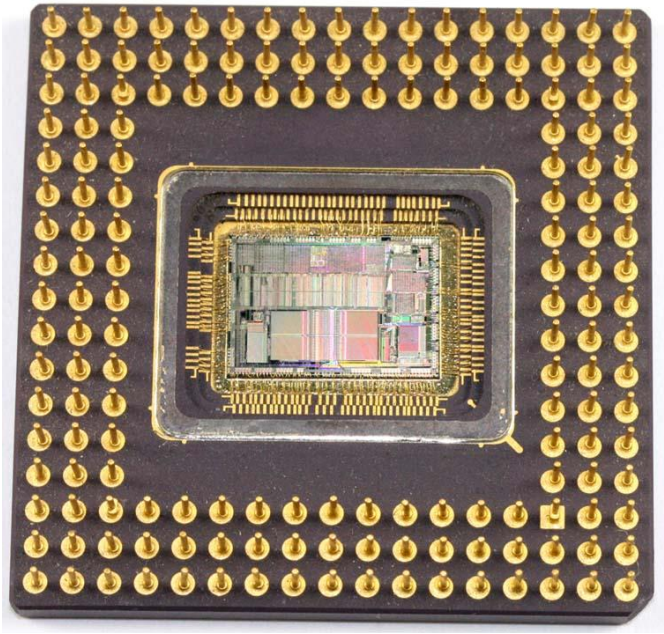
Gary Kildall

PL/M: **P**rogramming **L**anguage for **M**icrocomputers

The first (high level) programming language and first compiler specifically for microprocessors: PL/M. (1973)

```
FIND: PROCEDURE(PA,PB) BYTE;
    DECLARE (PA,PB) BYTE;
    /* FIND THE STRING IN SCRATCH STARTING AT PA AND ENDING AT PB */
    DECLARE J ADDRESS,
        (K, MATCH) BYTE;
    J = BACK ;
    MATCH = FALSE;
        DO WHILE NOT MATCH AND (MAXM > J);
        LAST,J = J + 1; /* START SCAN AT J */
        K = PA ; /* ATTEMPT STRING MATCH AT K */
            DO WHILE SCRATCH(K) = MEMORY(LAST) AND
                NOT (MATCH := K = PB);
            /* MATCHED ONE MORE CHARACTER */
            K = K + 1; LAST = LAST + 1;
            END;
        END;
    IF MATCH THEN /* MOVE STORAGE */
        DO; LAST = LAST - 1; CALL MOVER;
        END;
    RETURN MATCH;
    END FIND;
```

ALGOL(58,60,68): Algorithmic Language, 1958; FORTRAN, 1957; FORTRAN77, FORTRAN90; C, 1972-1973; BASIC, 1975



Intel 80486: 1.2M Transistors, 0.6um , 32-Bit, Clock Rate: 100MHz, April 1974, L1 Cache 16Kb

Pioneer Operating Systems

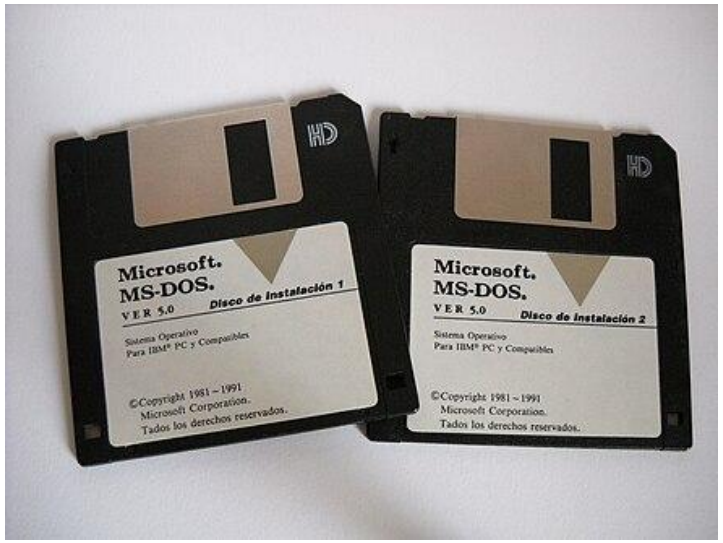
```
Loading CPM.SYS...

CP/M-86 for the IBM PC/XT/AT, Vers. 1.1 (Patched)
Copyright (C) 1983, Digital Research


Hardware Supported :

          Diskette Drive(s) : 3
         Hard Disk Drive(s) : 1
        Parallel Printer(s) : 1
             Serial Port(s) : 1
                Memory (Kb) : 640


D>a:
A>dir
A: PIP      CMD : STAT     CMD : SUBMIT   CMD : ASM86    CMD
A: GENCMD   CMD : DDT86    CMD : TOD      CMD : ED       CMD
A: HELP     CMD : HELP     HLP : SYS      CMD : ASSIGN   CMD
A: FORMAT   CMD : CLDIR    CMD : WRTLDR   CMD : BOOTPCDS SYS
A: BOOTWIN  SYS : CPM      H86 : WINSTALL SUB : PD       CMD
A: WCPM     SYS : DISKUTIL CMD
A>_
     User 0       0:00:11          Jan. 1, 2000
```

Pioneer Operating Systems

```
Starting MS-DOS...


HIMEM is testing extended memory...done.

C:\>C:\DOS\SMARTDRV.EXE /X

MODE prepare code page function completed

MODE select code page function completed
C:\>dir

 Volume in drive C is MS-DOS_6
 Volume Serial Number is 40B4-7F23
 Directory of C:\

DOS            <DIR>          12.05.20   15:57
COMMAND  COM        54 645 94.05.31    6:22
WINA20   386         9 349 94.05.31    6:22
CONFIG   SYS           144 12.05.20   15:57
AUTOEXEC BAT           188 12.05.20   15:57
        5 file(s)         64 326 bytes
                      24 760 320 bytes free

C:\>_
```